

26. In a bisexual flower, if androecium and gynoecium mature at different times, the phenomenon is known as :

- (a) Dichogamy (b) Herkogamy
(c) Heterogamy (d) Monogamy

I.A.S. (Pre) 2002

Ans. (a)

In a bisexual flower, if androecium and gynoecium mature at different times the phenomenon is known as dichogamy. It is of two types : (1) Protandry (2) Protogyny. Protogyny is a condition of flower in which female parts mature before the male ones, while protandry is the condition in which male part of flower mature before the female ones.

27. Which of the following genomic conditions is found in commercial edible wheat?

- (a) AB genome (b) ABC genome
(c) ABCD genome (d) ABD genome

Jharkhand P.C.S. (Pre) 2023

Ans. (d)

ABD genome (AABBDD) is found in commercial edible wheat (i.e., Bread wheat; *Triticum aestivum*).

28. How many chromosomes are found in the endosperm of maize?

- (a) 20 (b) 30
(c) 40 (d) 23

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Ans. (b)

In maize, meiocyte contains 20 chromosomes and gamete has 10 chromosomes. Endosperm is triploid in maize. Hence, the number of chromosomes in the endosperm (triploid tissue that nourishes embryo) of maize is 30.

29. Which component of plants receives stimulus for flowering?

- (a) Stems (b) Branches
(c) Leaves (d) Roots

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

The stimulus for flowering which are light, energy and carbon dioxide are received by the leaves of a plant. The leaves are the sites of food production for the flowering plant. Apart from this, florigen, the flowering hormone or stimulus is also produced in leaves.

V. Plants : Economic Importance

Notes

- The primary necessities of man are threefolds—food, clothing and shelter.
- The most essential need of man is food.
- The food primarily comes from the cereals (e.g. rice, wheat, maize, oat, barley), millets (e.g. sorghum, pearl millet), pulses, vegetables and fruits. For clothing plants are indispensable.
- The plants that yield fibres are second only to food plants. Besides above, coal, fuels, medicines and other useful substances are obtained from the plants.

Sunflower oil :

- It is the non-volatile oil which is obtained by crushing the seeds of sunflower (*Helianthus annuus*—Family-Asteraceae).
- Nutritionally, it is better than many other cooking mediums. It can be safely used by heart patients.
- Sunflower oil does not raise the cholesterol level in blood. High cholesterol is bad for heart.
- Sunflower oil contains proteins and vitamin A, D and E.
- The oil is easily digested.
- Being of semi-drying and stable type, sunflower oil is also used in making paint, varnish, and soap.

Essential Oils :

- The essential oils also known as volatile oils evaporates in contact with the air and possesses a pleasant fragrance.
 - Chemically the essential oils are very complex.
 - They are found in many different species of plants of various families.
 - Some essential oil yielding plants are as follows :
- Chandan (Sandalwood) :** It is a tree, a native of the highlands of South India and the Malayan Archipelago. The major demand for sandalwood oil is from the perfumery trade. It is also used medicinally.
 - Camphor Oil :** The camphor oil is obtained from the wood and leaves of camphor tree (*Cinnamomum camphora*—Family-Lauraceae). Camphor oil is mainly used in the manufacture of celluloid and various nitrocellulose compounds. They are widely used in the preparation of medicines and perfumes.
 - Mint Oil :** Mint oil is obtained from the leaves by distillation process of *Mentha arvensis*. Mint oil is the main source of menthol. It is used in the treatment of colds.

Indian Gooseberry–Amla (*Phyllanthus emblica*) :

- A common tree with edible fruits. The fruits are used in diarrhoea and dysentery. The bark, leaves and fruits are used in dyeing and tanning. The wood yields excellent charcoal. The pickle or jam is prepared from the fruits. The fruit is very rich in vitamin C.
- Highyield varieties of Amla are Kanchan, Krishna, Chakaiya, BSR1, Banarsi, NA7 etc.

Mango (*Mangifera indica*) :

- Mango is the most popular and the choicest fruits of India and occupies a prominent place among the best fruits of the world.
- The mango fruit is one of the most widely cultivated fruits in the tropics. It has a rich, luscious, aromatic flavour and a delicious taste in which sweetness and acidity are delightfully blended. Young and unripe fruits are used in pickles, chutney, amchur etc.
- The unripe fruit has citric, malic, oxalic and succinic acids. Ripe fruits constitute a rich source of vitamin A. The fruit is also a rich source of potassium.

Banana (*Musa paradisiaca*) :

- The fruits have a high content of carbohydrates with some proteins and fats. Their food value is three times that of wheat.

Guava (*Psidium guajava*) :

- The fruits are edible. It is aromatic, sweet, juicy and highly flavoured. It contains acid, sugar, and pectin.
- It is one of the richest sources of vitamins A, B and C. It is commonly used for making jellies, jams and paste.

Apple (*Malus domestica*) :

- Apples are valued mainly as dessert fruits. Apples are rich in pectin and are useful in diarrhoea. Apple juice, syrups and vinegar reduce curd tension of milk used in infant feeding.
- Apple murabba is regarded as a stimulant for the heart, it is reported to relieve physical heaviness and mental strain.
- The vitamins, salts and organic acids are concentrated particularly in and just below the skin and the fruit should be eaten unpeeled. Apple is considered a good source of potassium.
- The mineral constituents of apple are Ca, Mg, K, Na, P, Cl, S, and Fe.

Grapes (*Vitis vinifera*) :

- Grape is a delicious, refreshing and nourishing fruit. It is classed as a protective food, is easily digestible and has large quantities of sugar, minerals like calcium, iron and vitamin B.
- Grapes are also used in the preparation of wines.

Citrus Fruits :

- Citrus fruits contain considerable amounts of essential vitamin C and some other fruit acids. The fruits are used in the preparation of juices, squashes and other canned products.
- Citrus fruit intake is associated with a reduced risk of stomach cancer. Lemons have the highest concentration of citrate of any citrus fruit, and daily consumption of lemonade has been shown to decrease the rate of kidney stone formation.

Neem (*Azadirachta indica*) :

- Neem is also known as **village pharmacy**. The components extracted from the neem plant have been used in traditional medicines for the cure of multiple diseases including cancer for centuries. The extract of seeds, leaves, flowers, and fruits of neem have consistently shown chemopreventive and anti-tumor effects in different types of cancer.
- Azadirachtin and nimbolide are among the few bioactive components of neem. Nimbolide is a terpenoid lactone derived from neem tree that displays a variety of biological activities including anti-malarial and anti-cancer activity. Azadirachtin has a role as a hepatoprotective agent and it is also the active ingredient in many pesticides.
- Neem is used as fungicides and insecticides to control the fungal infection and insects in plant protection.
- Neem is a natural mosquito repellent which is 100% effective. Camphor is also used as a natural mosquito repellent.

Biodiesel :

- Biofuels made from plants could one day help us to reduce our dependency on fossil fuel.
- Ratanjot (*Jatropha curcas*), Karanja (*Pongamia pinnata*) and Neem (*Azadirachta indica*) are having great potential as a future oil source for biodiesel that does not compete with food. In biodiesel hydrocarbons and lipids are present in abundant amount.
- The first biodiesel plant of our country has been established at Kakinada (A.P.).
- Biofuel Ethanol is derived mainly from the sugarcane and maize.

Chilgoza pine (*Pinus geradiana*) :

- Pinus seed is called chilgoza.
- Chilgoza is rich with carbohydrates and protein.

Tulsi (*Ocimum tenuiflorum*) :

- Tulsi/Holy Basil is an aromatic perennial plant.
- Tulsi has been used for thousands of years in Ayurveda for its diverse healing properties. It has many medicinal properties. The leaves strengthen the stomach and help in respiratory diseases. It reduces kidney stone formation.

Haldi/Turmeric (*Curcuma longa*) :

- Turmeric has been used for thousands of years not only for cooking but also as a medicine due to its incredible healing properties.
- Turmeric has powerful anti-inflammatory properties and can help to improve symptoms of arthritis and joint pains.
- Turmeric is also useful in dementia and Alzheimer's disease. It is antioxidant and lowers cholesterol.
- It slows the spread of breast cancer.
- Turmeric powder used as a mouthwash in improving oral health.

Lemon :

- The fruit of lemon's health benefits are as follows :
 - (a) It promotes hydration.
 - (b) It is a good source of vitamin C.
 - (c) It supports weight loss.
 - (d) Enhance the immune system of the body.
 - (e) It aids in digestion.
 - (f) It helps to prevent kidney stone formation.

Brahmi/Waterhyssop (*Bacopa monnieri*) :

- Waterhyssop or herb of grace is a perennial creeping herb used in Ayurveda where it is also known as Brahmi.
- In Ayurveda, it is believed to sharpen the brain by protecting cells and increasing chemicals associated with learning and memory.
- It is used to treat Alzheimer's disease, anxiety and ADHD (Attention Deficit Hyperactivity Disorder-difficult for a person to pay attention and control impulsive behaviour) along with fighting stress and improving memory.

Pudina/Mint :

- Mint is a calming and soothing herb that has been used for thousands of years to aid with upset stomach or indigestion.
- It is thought to increase bile secretion and encourage bile flow, which helps to speed and ease digestion (and which may also support healthy cholesterol level).

Sadabahar or Vinca rosea (*Cantharanthus roseus*) :

- It is an evergreen shrub that works as an ornamental plant and for medicinal purposes. The smooth, glossy, and dark green colour leaves along with flowers are said to act as a natural medicine for type-2 diabetes.

Beverages :

- **Tea (*Camellia sinensis*) :** Tea is the most important non-alcoholic beverage; it is a safe and helpful stimulant. The main chemical ingredient of tea is caffeine (an alkaloid), which make a person feel refreshed.
- **Coffee :** Coffee is a brewed drink prepared from the roasted coffee beans. Its main chemical ingredients are caffeine, tannin, diterpenes, chlorogenic acid and melanoidins, which may affect human health.

- **Cocoa :** Cocoa is the dried and fully fermented seeds of **Theobroma cacao**, from which cocoa solids (a mixture of non-fat substances) and cocoa butter (the fat) is extracted. Cocoa beans are the basis of chocolate and used as a beverage.
- **Kasni (*Chicory-Chichorium intybus*):** It is a woody, perennial herbaceous plant, usually with bright blue flowers. The chicory coffee is a beverage made using the roots of the chicory plant.

Other important facts :

- The main source of iodine is the seaweeds (Algae).
- Vanilla is a flavouring substance derived from orchids.
- The main organic acids of tomato are citric acid and malic acid. Oxalic acid is also found in little amount.
- The flour of kuttu is obtained from the seeds of **Fagopyrum esculentum** which is enriched with carbohydrate and protein.
- Peppermint is obtained from the leaves of **Mentha piperita**. Peppermint oil has a high concentration of natural pesticides. Its leaves and oil have a cooling effect when used topically for muscle pain, nerve pain and relief from itching.
- **Quinine** is a medicine which is used to cure **malaria**. It is derived from the bark of **cinchona tree** which belongs to the family **Rubiaceae**. This plant is evergreen shrub or tree.
- **Arteether** is a potent antimalaria drug derived from **artemisinin**, a **sesquiterpene** lactone isolated from **Artemisia annua**. This medicine is used in severe malarial condition when other medicine are ineffective to control malaria.
- **Chloroquine, primaquine, atovaquone** are some other medicines used to treat malaria.
- **Cotton** is a soft, fluffy staple fibre that grows in a boll, or protective case, around the seeds of the cotton plants (**Gossypium Malvaceae**). Chemical composition of cotton is around as follows – Cellulose - 91%, Water - 7.85%, Protoplasm 0.55%; Fats - 0.40%, and Minerals - 0.20%.
- **Opium** is the dried latex obtained from the opium poppy (**Papaver somniferum**). Approximately 12% of the opium latex is made up of the **analgesic alkaloid morphine**, which is processed chemically to produce **heroin** and other synthetic **opioids** for medicinal use and for illegal drug trade. The latex also contains the closely related opiates **codeine** and **thebaine**, and non-analgesic alkaloids such as **papaverine** and **noscipine**. The morphine is obtained from the unripened fruits of the plant.

Question Bank

1. 'Kanchan' is an improved variety of :

- (a) Grapes (b) Indian gooseberry
(c) Guava (d) Mango

U.P. Lower Sub. (Pre) 2015

Ans. (b)

The fruits of Indian gooseberry are rich in medicinal properties. It is rich in vitamin C. Kanchan, Krishna and Banarasi are some of its improved varieties.

2. The hybrid variety of mango developed from the cross of Neelum and Alphonso is :

- (a) Amrapali (b) Dasherri
(c) Ratna (d) Badami

M.P. P.C.S. (Pre) 2020

Ans. (c)

Ratna is a mango variety of Indian origin grown in the state Maharashtra and some parts of Karnataka. It is a hybrid variant made from the Alphonso and Neelum mango. It is considered to be one of the high yielding mango varieties.

3. 'Kashi Lalima' is the variety of which of the following crops?

- (a) Carrot (b) Okra/ Lady finger
(c) Onion (d) Tomato

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

'Kashi Lalima' is the variety of okra/ladies' finger. It was developed by agricultural experts at the Indian Institute of Vegetable Research (IIVR) in Varanasi. This red colour variant is much more nutritious and healthy than the regular green okra due to its anti-oxidant properties and its richness in calcium and iron.

4. Co. 1148 is an important variety of :

- (a) Wheat (b) Cotton
(c) Maize (d) Sugarcane

U.P.P.C.S. (Mains) 2017

Ans. (d)

Co.1148 is an important variety of sugarcane for eastern tract of Uttar Pradesh. Its canes are medium thick. It is a very good germinator with profuse tillering. This variety is moderately resistant to drought.

5. Which of the following pairs is not correctly matched?

Crop		Variety
(a) Mustard	–	Varuna
(b) Field Pea	–	Sapna
(c) Linseed	–	Surya
(d) Groundnut	–	Kaushal

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

'Surya' is the name of the varieties of papaya, brinjal, mango, potato, tomato, etc. It is not a variety of linseed. Other pairs are correctly matched.

6. The botanical name of macaroni wheat is :

- (a) Triticum monococcum
(b) Triticum durum
(c) Triticum dicoccum
(d) Triticum aestivum

Jharkhand P.C.S. (Pre) 2021

Ans. (b)

The botanical name of macaroni (macaroni) wheat is Triticum durum. It is a tetraploid species of wheat. It is the second most cultivated species of wheat after common wheat, although it represents only about 5% to 10% of global wheat production.

7. Biologically, synthesized nano particles are used as a new eco-friendly technology, which has big possibility in cancer treatment. Extracts of a plant are used in this technique, this plant is :

- (a) Calotropis gigantea
(b) Parthenium hysterophorus
(c) Rice
(d) Lemongrass

U.P.P.C.S. (Mains) 2004

Ans. (b)

Leaf of Parthenium hysterophorus plant exhibits significant medicinal attributes including anti-cancer property.

8. Litmus, the acid-base indicator is obtained from –

- (a) Bacterium (b) Lichen
(c) Virus (d) None of the above

U.P.P.C.S. (Mains) 2010

Ans. (b)

Litmus is a water soluble mixture of different dyes extracted from lichens. It is often absorbed onto filter paper to produce one of the oldest forms of pH indicator, used to test materials for acidity and alkalinity.

9. With reference to neem tree, consider the following statements :

1. Neem oil can be used as a pesticide to control the proliferation of some species of insects and mites.
2. Neem seeds are used in the manufacture of bio-fuels and hospital detergents.
3. Neem oil has applications in pharmaceutical industry.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2014

Ans. (c)

Azadirachtin is a substance with pesticidal properties found in neem plant extracts such as 'neem oil'. Neem oil can be used as a pesticide to control the proliferation of some species of insect and mites. Neem oil, as a traditional medical remedy, is used as anti-bacterial, anti-fungal, insect repellent and treatment of skin diseases. Thus, neem oil has many applications in pharmaceutical industry. Neem Methyl Ester (Bio-diesel) can be prepared by a two-step process of esterification and transesterification from neem oil with methanol in the presence of a catalyst. Neem oil is used in the production of biodegradable detergents but there is no proper evidence for the use of neem seeds in the manufacture of hospital detergents. Hence, statements 1 and 3 are correct, while statement 2 is incorrect.

10. Pine, Fir, Spruce, Cedar, Larch and Cypress are the famous timber-yielding plants of which several also occur widely in the hilly regions of India. All these belong to :

- (a) angiosperms (b) gymnosperms
(c) monocotyledons (d) dicotyledons
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (b)

Pine, Fir, Spruce, Cedar, Larch and Cypress are the famous timber-producing plants. These are angular trees found in the mountain forests. All these belong to Clade gymnosperms. The term gymnosperm literally means 'naked seeds', which is based on the unenclosed condition of their seeds (called ovules in their unfertilized state). The non-encased condition of their seeds contrasts with the seeds and ovules of flowering plants (angiosperms), which are enclosed within an ovary.

11. Consider the following statements :

1. Some mushrooms have medicinal properties.
2. Some mushrooms have psychoactive properties.
3. Some mushrooms have insecticidal properties.
4. Some mushrooms have bioluminescent properties.

How many of the above statements are correct?

- (a) Only one (b) Only two
(c) Only three (d) All four

I.A.S. (Pre) 2023

Ans. (d)

Mushrooms, which have always been appreciated for their culinary and nutritional value, are now increasingly valued for their many important medicinal properties, so much so that they are used not only as dietary food (functional foods) but also in the form of dietary supplements, nutraceuticals, and mycotherapy products. Medicinal mushrooms (MMs) are reported to have numerous pharmacological actions such as antimicrobial, anti-inflammatory, immunomodulatory, antidiabetic, cytotoxic, antioxidant, hepatoprotective, anticancer, antioxidant, antiallergic, antihyperlipidemic, and prebiotic properties, among others. Medicinal mushrooms also represent a valuable natural source of biologically active compounds with fungicidal, herbicidal and insecticidal properties. Hence, statement 1 and 3 are correct.

Magic mushrooms, also known as psilocybin mushrooms or 'shrooms', have a long and rich history of use in various cultures worldwide for their psychoactive and spiritual properties. The active compounds in these mushrooms, psilocybin, and psilocin, can cause profound changes in perception, mood and thinking. These can induce hallucinogenic or psychedelic effects when consumed. Hence, statement 2 is correct.

Some mushrooms, such as Lions Mane, Ghost Fungus and Jack-O-Lantern, have bioluminescent properties, causing them to have a green glow. Certain mushrooms release a glow of cold light due to a chemical reaction between oxyluciferin molecules, an enzyme called luciferase, and oxygen. Hence, statement 4 is also correct.

12. Which of the following is a medicinal crop?

- (a) Sugarcane (b) Aloe vera
(c) Cotton (d) Mahua

M.P. P.C.S. (Pre) 2018

Ans. (b)

Aloe vera is extensively used in beauty products. It has got antiviral and antibacterial properties and the ability to help beat everything from constipation to diabetes. Sugarcane provides sugar, cotton provides natural fibres and mahua provide mahua flower used in food and preparation of wine and its seeds provide vegetable oil.

13. Tulsi plant has medicinal values due to the presence of the following :

- (a) Inorganic Acid (b) Organic Acid

- (c) Phenols and Flavonoids (d) None of the above

U.P.P.C.S. (Pre) 2019

U.P. R.O./A.R.O. (Pre) 2021

Ans. (c)

Ocimum tenuiflorum (synonym – Ocimum sanctum), also known as Tulsi or Holy basil is an aromatic plant and it belongs to the family Lamiaceae. In its leaf many important phytochemical components are present which possess antiseptic, analgesic, anti-inflammatory, antimicrobial and antioxidant properties. Various secondary metabolites such as flavonoids, saponins, glycoside, terpenoid and phenol etc. are present in tulsi leaf extract which increase its medicinal values.

14. With reference to 'palm oil', consider the following statements :

1. The palm oil tree is native to Southeast Asia.
2. The palm oil is a raw material for some industries producing lipstick and perfumes.
3. The palm oil can be used to produce biodiesel.

Which of the statements given above are correct?

- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (b)

Palm oil tree (Elaeis—from Greek 'oil') is a genus of palms containing two species, called oil palms. They are used in commercial agriculture in the production of palm oil. The African oil palm *Elaeis guineensis* is the principal source of palm oil. It is native to west and southwest Africa, occurring between Angola and Gambia. The American oil palm *Elaeis oleifera* is native to tropical Central and South America, and is used locally for oil production. Hence, statement 1 is incorrect.

Since palm oil contains more saturated fats than oils made from canola, corn, linseed, soyabeans, safflower, and sunflowers, it can withstand extreme deep-frying heat and resists oxidation. It contains no trans fat, and its use in food has increased.

Palm oil is cheap and versatile. It is found in everything from lipstick to ice cream. Palm oil is used in cosmetics for its moisturising and texturising properties. Palm oil is used in lipstick as it holds color well, doesn't melt at high temperatures, and has a smooth application and virtually no taste. Also, palm oil makes perfumes and deodorants more effective and helps the fragrance to last longer. Oil from *Elaeis guineensis* is also used as biofuel. Hence, statements 2 and 3 are correct.

15. According to India's National Policy on Biofuels, which of the following can be used as raw materials for the production of biofuels?

- | | |
|--------------------|-------------------------|
| 1. Cassava | 2. Damaged wheat grains |
| 3. Groundnut seeds | 4. Horse gram |
| 5. Rotten potatoes | 6. Sugar beet |

Select the correct answer using the code given below:

- (a) 1, 2, 5 and 6 only (b) 1, 3, 4 and 6 only
(c) 2, 3, 4 and 5 only (d) 1, 2, 3, 4, 5 and 6

I.A.S. (Pre) 2020

Ans. (a)

As per India's National Policy on Biofuels-2018, potential domestic raw materials for production of biofuels in the country are :

(i) For Ethanol Production: B-Molasses, sugarcane juice, biomass in form of grass, agriculture residues (rice straw, cotton stalk, corn cobs, sawdust, bagasse, etc.), sugar containing materials like **sugar beet**, sweet sorghum, etc. and starch containing materials such as corn, **cassava, rotten potatoes**, etc; **Damaged food grains like wheat**, broken rice, etc. which are unfit for human consumption; Food grains during surplus phase. Algal feedstock and cultivation of sea weeds can also be a potential feedstock for ethanol production.

(ii) For Biodiesel Production: Non-edible Oilseeds, Used Cooking Oil (UCO), Animal tallow, Acid Oil, Algal feedstock etc.

(iii) For Advanced Biofuels: Biomass, MSW, Industrial waste, Plastic waste etc.

The groundnut seeds or horse gram is not mentioned in this policy document. Hence, option (a) is the correct answer.

16. Canola refers to a special type of oilseed mustard varieties bred for human consumption. The main characteristic of these varieties is that the :

- (a) seeds have very high oil content
(b) oil is rich in unsaturated fatty acids
(c) oil has long shelf-life
(d) oil has very low Erucic acid content

I.A.S. (Pre) 2000

Ans. (d)

Canola refers to a special type of edible oil which was bred naturally from rapeseed at the University of Manitoba, Canada by Keith Downey and Baldur R. Stefansson in the early 1970s. Canola seeds are used to produce edible oil, which is considered as safe for human consumption because it has very low Erucic acid content. Canola oil has low saturated fat and contains both omega-6 and omega-3 fatty acid. According to USDA, canola oil contains about 7% saturated fat, 64% monounsaturated fat and 28% polyunsaturated fat.

17. Consider the following statements :

1. Moringa (drumstick tree) is a leguminous evergreen tree.
2. Tamarind tree is endemic to South Asia.
3. In India, most of the tamarind is collected as minor forest produce.
4. India exports tamarind and seeds of moringa.
5. Seeds of moringa and tamarind can be used in the production of biofuels.

Which of the statements given above are correct?

- (a) 1, 2, 4 and 5 (b) 3, 4 and 5
(c) 1, 3 and 4 (d) 1, 2, 3 and 5

I.A.S. (Pre) 2021

Ans. (b)

Moringa oleifera (drumstick tree), native to the Indian subcontinent, grows in the tropical and subtropical regions of the world. It is commonly known as 'drumstick tree' or 'horseradish tree'. Moringa can withstand both severe drought and mild frost conditions and hence widely cultivated across the world. It is not a leguminous tree as it does not fix nitrogen. It is widely cultivated for its young seed pods and leaves, used as vegetables and for traditional herbal medicine. Moringa and its value-added products have a huge market in North America, Europe, Gulf Countries and in China. India is the largest moringa exporter and meets 80 percent of its world demand. While moringa latte, tea and moringa capsules have huge demand in North America for its weight loss properties, recipes made using moringa oil and seeds are very popular among Chinese men as an aphrodisiac. In 2020-21, China procured huge quantity of moringa seeds from India. Seeds of Moringa can also be used in the production of biofuel. Tamarind is a hardwood tree, known scientifically as Tamarindus indica. It is native to Africa but also grows in India, Pakistan, and many other tropical regions. In India most of the tamarind is collected as minor forest produce. Tamarind is also actively exported from India. Biodiesel is derived from the tamarind seeds through the transesterification process. Hence, statements 1 and 2 are incorrect while other three statements are correct.

18. Consider the following:

1. Cashew
2. Papaya
3. Red sanders

How many of the above trees are actually native to India?

- (a) Only one (b) Only two

(c) All three

(d) None

I.A.S. (Pre) 2024

Ans. (a)

Cashew (*Anacardium occidentale* L.), a native of Brazil introduced to India just as other commercial crops like Rubber, Coffee, Tea etc. by the Portuguese nearly five centuries back. The first introduction of cashew in India was made in Goa from where it spread to other parts of the country. Native to tropical America, papaya (*Carica papaya*) originates from southern Mexico and Central America. It found its way to the Indian subcontinent centuries ago, likely through trade routes. At present, India leads the world in papaya production. Red Sanders or Red Sandalwood (*Pterocarpus santalinus*), is an Indian endemic tree species, with a restricted geographical range in the Eastern Ghats. The species is native to a distinct tract of forests in Andhra Pradesh. It is in 'endangered' category in the IUCN's Red List. Following the 2022 amendment to the Wild Life (Protection) Act 1972 (which came into force from 1.4.2023), Red Sanders is now listed in Schedule IV of the Act. Under the foreign trade policy of India, the import of Red Sanders is prohibited, while export is restricted. Despite the regulation and legal protection, illegal logging and timber extraction remains major conservation concern for the species. Hence, option (a) is the correct answer.

19. The fourth generation biofuels are obtained from genetically modified :

- (a) Archaeobacteria
(b) Starch yielding crops
(c) Ligno-cellulose yielding crops
(d) Algae

Rajasthan P.C.S. (Pre) 2024

Ans. (d)

Fourth generation biofuels are made using genetically modified (GM) microorganisms, like algae, yeast, fungus and cyanobacteria. They are produced by using these microorganisms to photosynthesize carbon dioxide into fuel. Although GM algae biofuel is a well-known alternative to fossil fuels, the potential environmental and health-related risks are still of great concern.

20. Which of the following leguminous plant is also a petro-plant?

- (a) Pigeon-pea (b) Pea
(c) Gram (d) Pongam

U.P.P.C.S. (Pre) 2010

Ans. (d)

The 'Pongam Tree' is known as one of the richest and brightest trees of India. Its scientific name is 'Pongamia pinnata'. A red oil obtained from the seeds of the 'Pongam Tree' can be used in embrocating the skin disease. It is used as bio-diesel as well.

21. The plants commonly known as (Petro-crops), they are rich in –

- | | |
|------------------|----------------|
| 1. Carbohydrates | 2. Hydrocarbon |
| 3. Protein | 4. Lipids |

Code :

- | | |
|----------------|----------------|
| (a) 1, 2 and 3 | (b) 2 and 4 |
| (c) 1, 3 and 4 | (d) 2, 3 and 4 |

U.P.P.C.S. (Mains) 2009

Ans. (b)

Petro-crops refer to those crops which yield biofuel compounds like bio-diesel. Petro-crops are rich in hydrocarbons and lipids. Jatropha is the genus of a flowering plant which is also known as petro-crop.

22. The first bio-diesel plant of the country is being established in –

- | | |
|--------------------|--------------------|
| (a) Andhra Pradesh | (b) Madhya Pradesh |
| (c) Maharashtra | (d) Tamil Nadu |

U.P.P.C.S. (Mains) Spl. 2004

Ans. (a)

India's first biodiesel plant was established at Kakinada in Andhra Pradesh.

23. Which of the following plant is used for 'bio-diesel' –

- | | |
|---------------|---------------------|
| (a) Sugarcane | (b) White carrot |
| (c) Senna | (d) Jatropha curcas |

Chhattisgarh P.C.S. (Pre) 2005

Ans. (d)

Jatropha Curcas, the plant is considered a future feedstock for biodiesel production because it is easily grown in a harsh environment and is a non-edible crop. It is an Angiospermic plant. At the time of cultivation, it needs very less amount of water compared to other plants. The seeds of these plant are used as biodiesel. Biodiesel has become more attractive because of its environmental benefits and is obtained from renewable resources.

24. Which one of the following is a bio-diesel plant ?

- | | |
|--------------|-----------------|
| (a) Guggal | (b) Rosha grass |
| (c) Jatropha | (d) Java grass |

U.P.P.C.S. (Mains) 2004

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

See the explanation of above question.

25. Which plant is known as bio-diesel plant?

- | | |
|----------------|----------------|
| (a) Isabgol | (b) Belladonna |
| (c) Ezadirecta | (d) Jatropha |

M.P.P.C.S. (Pre) 2006

Ans. (d)

See the explanation of above question.

26. In the context of alternative sources of energy, ethanol as a viable bio-fuel can be obtained from :

- | | |
|---------------|-----------|
| (a) Potato | (b) Rice |
| (c) Sugarcane | (d) Wheat |

I.A.S. (Pre) 2009

Ans. (c)

Sugarcane ethanol is an alcohol-based fuel produced by the fermentation of sugarcane juice and molasses. Because it is a clean, affordable and low-carbon biofuel, sugarcane ethanol has emerged as a leading renewable fuel for the transportation sector.

27. Molasses is an excellent raw material for :

- | | |
|-------------------|---------------|
| (a) Acetic acid | (b) Glycerine |
| (c) Power alcohol | (d) Urea |

Uttarakhand P.C.S. (Pre) 2002

Ans. (c)

Molasses is an excellent raw material for power alcohol. These molasses are prepared by sugarcane's juice.

28. We obtain chilgoza from the seed of species –

- | | |
|-----------|-----------|
| (a) Pine | (b) Palm |
| (c) Cycas | (d) Cedar |

U.P.P.C.S. (Pre) 2009

Ans. (a)

Pinus gerardiana is the biological name of chilgoza pine. Chilgoza is rich in carbohydrate and protein and it is obtained from pine seeds.

29. The chief mosquito repellent is obtained from –

- | | |
|-------------------|----------------------|
| (a) Basil (Tulsi) | (b) Lemon (Nibu) |
| (c) Margo (Neem) | (d) Turmeric (Haldi) |

U.P.P.C.S. (Mains) 2004

Ans. (c)

The chief mosquito repellent is obtained from neem. Neem has medicinal quality, medicines are prepared by its seeds and leaves. Basil and turmeric have the antibiotic quality. Lemon is the main source of vitamin C.

30. Match List-I and List-II and select the correct answer from the codes given the lists :

List-I (Plant)	List-II (Use)
A. Brahmi	1. Anti-diabetic
B. Sadabahar	2. Carminative
C. Pudina	3. Expectorant
D. Tulsi	4. Brain Tonic

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	1	2	3
(c)	3	2	1	4
(d)	2	4	3	1

U.P.P.C.S. (Mains) 2004

U.P.P.C.S. (Mains) 2008

Ans. (b)

Brahmi is also known as jalnimb. It commonly grows in marshy areas. It is a nervine tonic or brain tonic which acts as an anxiety agent and is used in several mental disorders. Pudina is a digestive ingredient. It promotes digestion, relieves from gas trouble (carminative), fights aches and pains and helps to relieve from a cough and cold. Tulsi, also known as 'Ocimum tenuiflorum', is an important medicinal plant, with immense benefits right from clear skin to dissolving kidney stones, it also helps in removing cough or it behaves like an expectorant agent. Sadabahar is evergreen every season. It has been found useful in treating diabetes.

31. Soft drinks such as colas contain significant quantities of :

- (a) Caffeine (b) Nicotine
(c) Tannin (d) Renin

I.A.S. (Pre) 2000

Ans. (a)

Soft drinks such as colas contain significant quantities of caffeine. Caffeine is a xanthine alkaloid compound that acts as a stimulant, helping to increase alertness and reduce fatigue. In colas and other caffeinated soft drinks, caffeine is added as an ingredient during the manufacturing process, while in beverages such as coffee and tea, caffeine is the natural ingredient.

32. The alkaloid caffeine is present in :

- (a) Tea only (b) Coffee only
(c) Tea and coffee both (d) Lemon water

U.P. P.C.S. (Pre) 2016

Ans. (c)

Caffeine is a naturally occurring compound found in the leaves and fruits of certain plants. It is found in coffee, tea, cocoa, cola, soft drinks. As a stimulant, caffeine acts on the brain and nervous system. In small doses, it can make anyone feel refreshed and focused. While in large doses, it can make to feel anxious and have difficulty in sleeping.

33. The most important stimulant present in tea leaves is

- (a) Caffeine (b) Phenylalanine
(c) Brucine (d) Theobromine

U.P. P.C.S. (Mains) 2017

Ans. (a)

Caffeine is the world's most widely consumed psychoactive drug or Central Nervous System (CNS) stimulant. It is mainly derived from the tea leaves and coffee beans.

34. Seaweeds are an important source of which one of the following?

- (a) Sulphur (b) Chlorine
(c) Bromine (d) Iodine

U.P.P.C.S. (Mains) 2015

U.P. R.O./A.R.O. (Pre) 2017

Ans. (d)

Seaweed is a type of algae that grows in salt water and (like terrestrial plants) needs sunlight to thrive. There are over ten thousand varieties of seaweed, many of which are good to eat. Seaweeds are best known as the richest natural source of iodine.

35. Consider the following :

1. Camphor 2. Chicory 3. Vanilla

Which of the above is/are plant product(s)?

- (a) 1 and 2 (b) 3 only
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2009

Ans. (d)

Camphor is a waxy, flammable, white or transparent solid with a strong aromatic odour. It is found in the wood of the camphor laurel (*Cinnamomum camphora*), a large evergreen tree found in Asia (particularly in Sumatra, Indonesia and Borneo) and also from the unrelated Kapur tree, a tall timber tree from the same region. Chicory is a somewhat woody, perennial, herbaceous plant usually with bright blue flowers, rarely white or pink. Its many varieties are cultivated for salad leaves and its root powder is used to reduce the bitterness of coffee. Vanilla is a flavour derived from orchids of the genus *Vanilla*, primarily from the Mexican species, flat-leaved vanilla.

36. 'Kuttu' flour is obtained from –

- (a) Tapioca (b) Fagopyrum
(c) Plantago (d) Eleusine

U.P.P.C.S. (Mains) 2010

Ans. (b)

Kuttu flour is obtained from the plant of Buckwheat (*Fagopyrum esculentum*). This plant is cultivated for its grain-like seeds and also used as a cover crop. Despite

the name, buckwheat is not related to wheat, as it is not a grass. Instead, buckwheat is related to sorrel, knotweed and rhubarb. Because its seeds are eaten, it is referred to as a pseudocereal.

37. In making the saffron spice, which one of the following parts of the plant is used?

- (a) Leaf (b) Petal
(c) Sepal (d) Stigma

I.A.S. (Pre) 2009

Ans. (d)

Saffron is a spice derived from the flower of *Crocus sativus*, commonly known as the 'saffron crocus'. Saffron crocus grows to about 20–30 cm (8–12 inch) and bears up to four flowers, each with three vivid crimson stigmas, which are the distal end of a carpel. The styles and stigmas, called threads are collected and dried to be used mainly as a seasoning and colouring agent in food. Saffron, is among the world's most costly spices by weight.

38. Which of the following part in peppermint has maximum percentage of oil ?

- (a) Root (b) Stem
(c) Leaf (d) Flower

U.P.P.C.S. (Pre) 2009

Ans. (c)

Peppermint oil or menthol oil is made up of peppermint leaves by steam distillation process. Its scientific name is *Mentha arvensis*. The oil has a high concentration of natural pesticides. Its main constituents are menthol (40.7%) and menthone (23.4%).

39. Which of the following substances is obtained from a tree?

- (a) Asafoetida (b) Bitumen
(c) Tincal (d) None of the above

U.P.P.C.S. (Pre) 2020

Ans. (a)

Asafoetida (Heeng) is the dried latex (oleo-gum resin), obtained from several species of *Ferula* which are herbaceous plants of the Umbelliferae family. It is extracted from the rhizome and fleshy tap roots. This spice is used as digestive aid, in food as a condiment, and in pickling.

40. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I (Product)	List-II (Source)
A. Opium	1. Bark
B. Hing	2. Root
C. Rubber	3. Fruit
D. Quinine	4. Stem

Code :

A	B	C	D
(a) 1	2	3	4
(b) 3	2	4	1
(c) 4	2	1	3
(d) 2	4	3	1

U.P.P.C.S. (Pre) 2017

Ans. (b)

The correct matching is as follows :

Opium	–	Fruit
Hing	–	Root
Rubber	–	Stem
Quinine	–	Bark

Opium is obtained from milky latex of *Papaver somniferum*. Hing is obtained mainly from the roots of a plant of *Ferula* species. Rubber is obtained from the liquid stuff turned out from the stem of plant and quinine is obtained from the bark of cinchona tree.

41. Asafoetida is obtained from :

- (a) Exudation from the stem (b) Extraction from roots
(c) Extraction from fruits (d) Extraction from leaves

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016

Ans. (a) & (b)

Asafoetida (Hing) is the dried latex (resin-like gum), obtained from several species of *Ferula* which are herbaceous plants of the Umbelliferae family. The oleo-gum resin comes from the dried sap extracted from the rhizome (modified stem) and fleshy tap roots and is used as spice. Asafoetida is endemic to Afghanistan and Iran. UPPSC had answered option (b) for this question in their first answer key.

42. Quinine which is a major drug for malaria is obtained from –

- (a) Angiosperm (b) Bacteria
(c) Yeast (d) Algae

R.A.S./R.T.S. (Pre) 1996

Ans. (a)

Quinine is a major drug used to treat malaria. Quinine is extracted from the bark of cinchona tree. It is an angiosperm plant. Quinine is a white crystalline alkaloid and bitter in taste. It is also used in the treatment of arthritis and prion-related disease.

43. Quinine, the effective medicine in the treatment of malaria, is extracted from :

- (a) Bark of Cinnamon (b) Bark of Cinchona
(c) Leaves of Ocimum (d) Bark of Acacia Catechu

U.P. P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

44. Quinine is a common drug for malaria that is obtained from a plant. Which part of the plant yields this drug?

- (a) Leaves (b) Root
(c) Stem bark (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

See the explanation of above question.

45. Quinine, used to treat malaria is obtained from which plant's part?

- (a) Leaf (b) Root
(c) Bark (d) Flower

Uttarakhand U.D.A./L.D.A. (Pre) 2007

Uttarakhand P.C.S. (Pre) 2002

Ans. (c)

See the explanation of above question.

46. In which part of the cinchona plant quinine occurs naturally?

- (a) Root (b) Fruit
(c) Seed (d) Bark

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (d)

See the explanation of above question.

47. The Arteether medicine which is used to cure malaria is obtained from a –

- (a) Seeded plant (b) Fungus
(c) Bacteria (d) Algae

U.P. P.C.S. (Pre) 2005

Ans. (a)

The Arteether medicine which is used to cure malaria is obtained from a seeded plant, Artemisia annua. Chloroquine, atovaquine, barium and paludrine medicines are also used to cure malaria. It is notable that malaria is caused by the Plasmodium parasite. The parasite can be spread to humans through the bites of infected female Anopheles mosquito.

48. The drug obtained from the bark of cinchona was used for treating malaria. The synthetic drug which replaced the natural product is :

- (a) Chloromycetin (b) Chloroquine
(c) Tetracycline (d) Ampicillin

U.P.P.C.S. (Pre) 2000

Ans. (b)

Quinine occurs naturally from the bark of the cinchona tree and was used in the treatment of malaria but now it is substituted by Chloroquine. It is a synthetic drug used to kill sensitive malaria parasites.

49. Hydroxychloroquine drug used for the treatment of CORONA patients is obtained from the bark of which of the following plants?

- (a) Aswagandha (b) Sarpagandha

(c) Cinchona

(d) Aloe vera

U.P. R.O./A.R.O. (Mains) 2016

Ans. (*)

Quinine is extracted from the bark of cinchona tree, which is used to treat malaria, arthritis and prion-related diseases. For the treatment of malaria, chloroquine and hydroxychloroquine are nowadays widely used instead of quinine, both of which are synthetic drugs. Thus, this question is wrong as hydroxychloroquine is not obtained from the plants. It is not the same as quinine, which is a naturally-occurring compound. Hydroxychloroquine is being studied to prevent and treat COVID-19, but all clinical trials conducted during 2020 found it is ineffective and may cause dangerous side effects.

50. Cork is obtained from which of the following plant?

- (a) Dalbergia (b) Cedrus
(c) Quercus (d) Argemone

Jharkhand P.C.S. (Pre) 2003

Ans. (c)

Cork is obtained from the bark of the oak tree, whose botanical name is Quercus suber. It is native to the Mediterranean region. Cork consists of irregularly shaped, thin-walled, waxy cells.

51. Major component of cotton is :

- (a) Protein (b) Fatty acid
(c) Cellulose (d) Glycerene

U.P. Lower Sub. (Pre) 2009

Ans. (c)

The chemical composition of cotton is around as follows–

Cellulose	–	91.00%
Water	–	7.85%
Protoplasm	–	0.55%
Elephatic elements	–	0.40%
Mineral salts	–	0.20%

52. Noscapine is obtained from :

- (a) Poppy (b) Basii
(c) Eucalyptus (d) Ephedra

U.P.P.C.S. (Mains) 2012

Ans. (a)

Noscapine is obtained from poppy.

53. Heroin is obtained from :

- (a) Indian hemp (b) Opium poppy
(c) Tobacco (d) Areca nut

U.P.P.C.S. (Pre) 2005

Ans. (b)

Heroin is obtained from the opium poppy (Papaver somniferum). Opium latex contains approximately 12% of the analgesic alkaloid morphine, which is processed chemically to produce heroin and other synthetic opioids for medicinal use and for the illegal drug trade.

54. We get morphine from :

- (a) Flower (b) Leave

(c) Fruit

(d) Stem

R.A.S./R.T.S. (Pre) 1994

Ans. (c)

Morphine is acquired from the unripe fruits of poppy straw. Poppy plant is an angiosperm plant from which opium and heroin are also acquired. The word morphine originates from Morpheus and it is used as a painkiller medicine and sleeping pills. Morphine directly affects central nervous system.

55. Opium is obtained from :

- (a) Dried leaves (b) Roots
(c) Latex from unripe fruit (d) Latex from ripe fruit

U.P.R.O./A.R.O. (Mains) 2014

Ans. (c)

Opium is the dried latex. The dried latex is obtained from the unripe fruit, by making shallow cuts.

56. Opium is obtained from which part of the poppy plant ?

- (a) Flower (b) Leaves
(c) Unripe Fruits (d) Roots

U.P.P.C.S. (Mains) 2011

Ans. (c)

See the explanation of above question.

57. The main constituent of opium is :

- (a) Morphine (b) Heroin
(c) Atropine (d) Quinine

U.P. P.C.S. (Mains) 2016

Ans. (a)

Opium is a substance that is derived from the collecting and drying processes of the milky juice that comes from the unripe fruit of the poppy plant. The primary component of opium is 12% morphine which is an alkaloid that is often processed chemically to produce illegal drugs such as heroin.

58. What is the botanical name of opium?

- (a) Emblica officinalis (b) Papaver Somniferum
(c) Rauwolfia Serpentina (d) Cinchona Species

M.P.P.C.S. (Pre) 2017

Ans. (b)

Papaver somniferum is a species of flowering plants in the family Papaveraceae. Its fruit has many seeded capsules which are scattered by the hole or valve. Opium is found from latex milk of its immature capsules. Many alkaloids like morphine, codeine, papaverine are obtained from opium.

59. Opium is of which category medicine ?

- (a) Depressant (b) Stimulant
(c) Hallucinogenic (d) None of the above

M.P.P.C.S. (Pre) 2019

Ans. (a)

Opium is a depressant (sedative) narcotic drug. It slows down body functions like heart rate and breathing. It also reduces pain and has been used to synthesize a very wide range of painkilling drugs. The opium latex is about 12% morphine, which is the opiate that is used to produce codeine, heroin and other synthetic opioids.

60. Which among the following is a Narcotic analgesic drug?

- (a) Morphine (b) Arsphenamine
(c) Aspirin (d) Paracetamol

Rajasthan P.C.S. (Pre) 2023

Ans. (a)

Opioids (narcotic analgesic) are a class of medicines that are used to provide relief from moderate-to-severe acute or chronic pain. Among the given options, morphine is a narcotic analgesic drug which is a natural alkaloid of opium. Morphine is used to treat pain severe enough to require daily, around-the-clock, long-term opioid treatment and when other pain medicines did not work well enough or cannot be tolerated. Morphine acts directly on the central nervous system (CNS) to relieve pain but has a high potential for addiction, with tolerance and both physical and psychological dependence developing rapidly.

61. Tapa cloth is obtained from the plants of which one of the following family?

- (a) Asclepiadaceae (b) Moraceae
(c) Graminae (d) Malvaceae

U.P. P.C.S. (Pre) 2021

Ans. (b)

Bark cloth, or tapa, is not a woven material, but made from bark that has been softened through a process of soaking and beating. The inner bark is taken from several types of trees or shrubs, Moraceae often mulberry and fig, and designs are applied with paints and vegetable dyes of light brown, red, and black. Tapa cloth is made in the islands of the Pacific Ocean, primarily in Samoa, Toga and Fiji, but as far afield as Nine, Cook Islands, Solomon Islands, Vanuatu, New Zealand, Papua New Guinea and Hawaii (where it is called 'Kapa').

62. Iodine is obtained from –

- (a) Laminaria (b) Polysiphonia
(c) Chara (d) Spirogyra

U.P.P.C.S. (Mains) 2004

Ans. (a)

Laminaria is a brown algae commonly called 'kelp'. Laminaria is a type of seaweed, native to Japan. It is abundantly available in the sea and also found on land. Laminaria contains iodine. It is also a rich source of iron and potassium. Despite serious safety concerns about laminaria, some people use laminaria as medicine.

63. Which one of the following is a substance abundantly available in the sea and administered in a certain deficiency disease?

- (a) Iron (b) Vitamin A
(c) Fluorine (d) Iodine

U.P.P.C.S (Pre) 2010

Ans. (d)

See the explanation of above question.